Isogeny and torsion fiber next gates

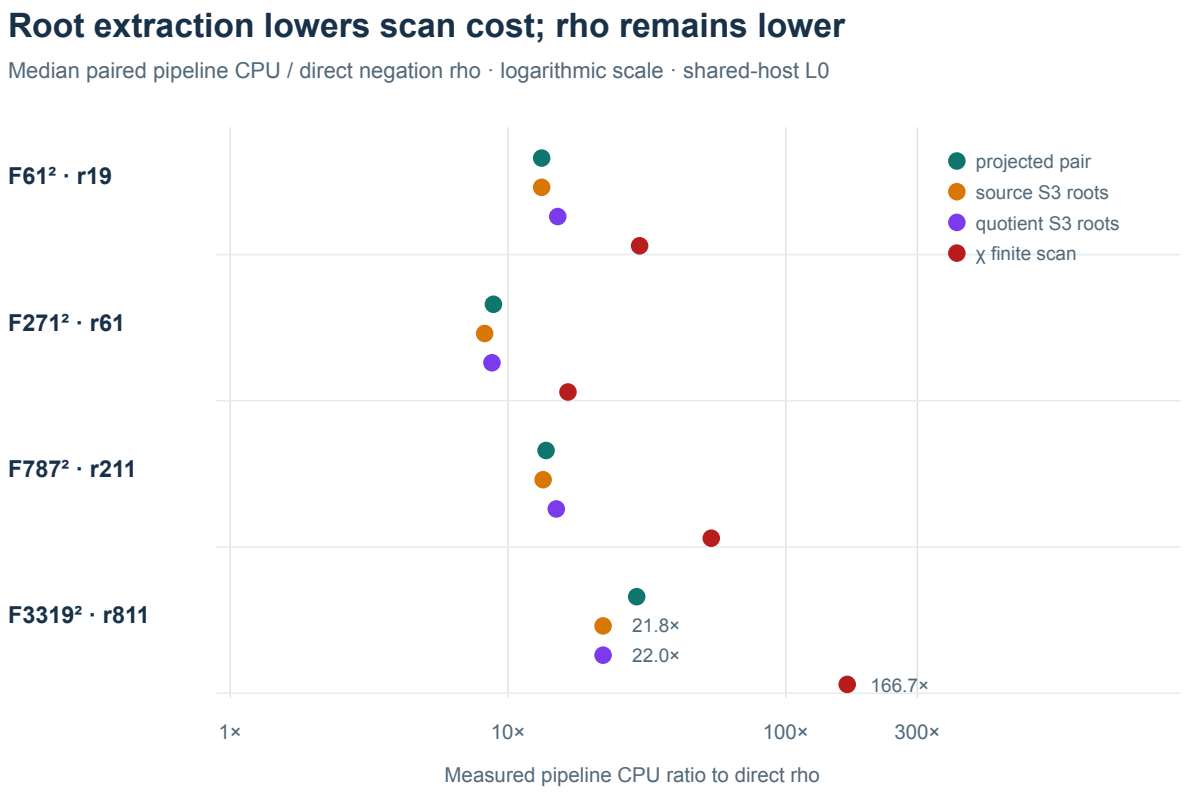
Results report, 2026-10-15. Native Rust, public frozen targets, shared-host L0.

The finite χ polynomial scan checked 22/22 exhaustive cases, solved 12/12 paired relation rounds, and independently replayed 99 scalar records across nine arms. The later direct source S3 and quotient-root controls each matched the independent signed-pair sets on 22/22 smallest-cell targets and solved 12/12 paired rounds. Their 11-arm run independently replayed 123/132 scalar outputs; nine raw-pair rows exhausted their cap. Projected-pair, χ finite-scan, source-root and quotient-root relation counts, rank and trials were identical in every paired round. The direct source-root/negation-rho paired pipeline CPU ratios were 15.474, 6.014, 14.669 and 30.631 on the four frozen cells. These pipeline CPU costs exclude common input reading and calibration; the portable counted S conversion is incomplete.

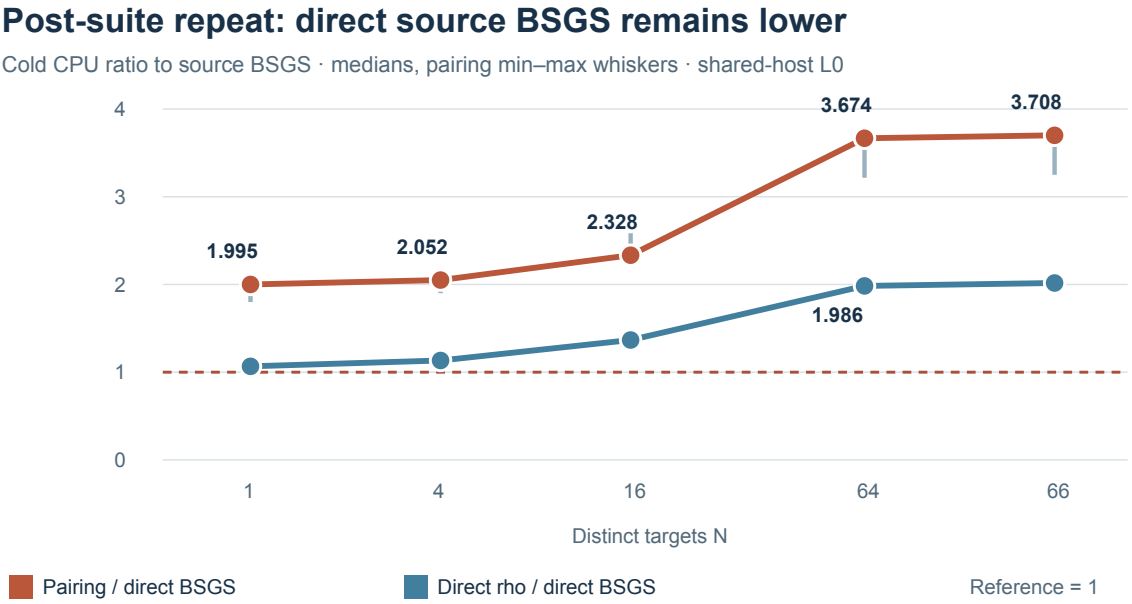
A separately frozen V3 repeat removed four answer fields before the solver run. The solver rejected an answer-bearing input and used a framework path without a planted answer. Its source and quotient roots again matched 22/22 exhaustive target sets and each solved 12/12 paired rounds; the separate checker replayed 123/132 scalar rows. All 132 normalized relation/recovery rows matched V2 exactly. Paired source-root/rho CPU ratios were 13.237, 8.254, 13.427 and 21.844; paired finite-scan/source-root ratios were 2.125, 1.988, 3.864 and 7.571. Native preprocessing and the independent checker read the private fixture; the solver process read only the public one.

The controlled post-V3 release library suite passed 4,083 tests, with zero failures and 117 ignored; its stdout and stderr are retained in the study directory. The touched examples compile, and exact native runs and independent replays provide their substantive checks.

An explicit degree-three map over $F_{(67^2)}$ links curves with distinct j values, 7 and 11. Source, transported-image and destination-direct relation arms reached rank nine and recovered scalar 41 in three rounds; 15/15 arm-round values passed independent replay. An exhaustive audit mapped 4575 source points onto 1525 image points, with exactly three rational preimages each. All 61 target-subgroup images were distinct and 3721 ordered subgroup-pair homomorphism checks passed. Lean proves conditional subgroup injectivity and relation reflection; the concrete curve/map and S3 polynomial bridges remain open.



At 64 distinct order-67 targets over F401, the post-suite V7B run priced cold direct source-group BSGS against direct prime-field rho and pairing with reused H and a finite-field BSGS table. Median complete CPU was 0.378, 0.938 and 1.219 ms, respectively. Paired pairing/source-BSGS ratios were 3.224, 3.696 and 3.674; their median was 3.674. Independent arithmetic replayed all 594 scalar outputs. Source BSGS was lower at every prefix in all three rounds. V7, measured with release tests running, had a median ratio of 5.190 at 64 targets; the difference is shared-host timing variation.



These results leave the direct χ-coordinate resultant, a full-image destination factor base, portable counted primitive conversion, isolated increasing-order comparisons, independent-host replay, and the concrete Lean curve bridges open. The study run reports contain frozen specs, hashes, raw receipts, failed attempts and exact replay. The canonical ECDLP frontier and progress charts were checked and left unchanged because no counted-S crossover was admitted.